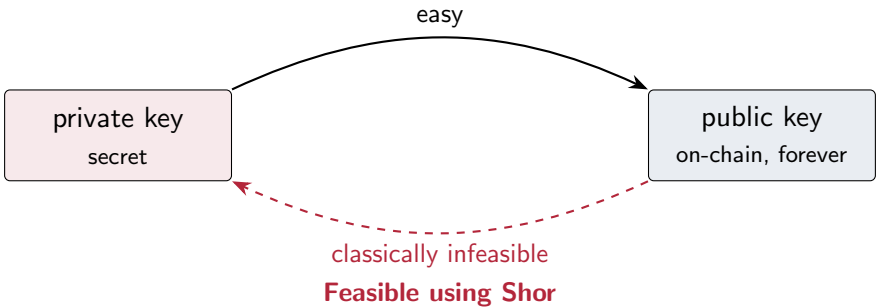
Post-Quantum Blockchains

Migration is hard

Brett Falk

University of Pennsylvania

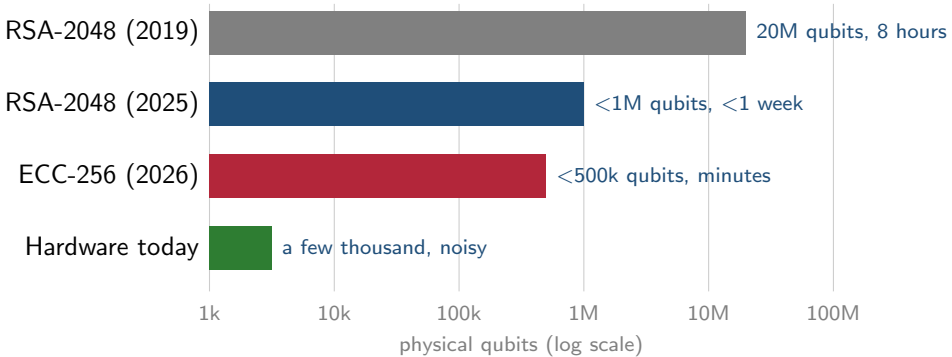
A quantum attacker can extract your private key



This breaks every signature scheme blockchains use today:

ECDSA, **EdDSA**, **Schnorr**, **BLS**

The estimated cost of the attack keeps falling



Quantum frontiers may be closer than they appear

Mar 25, 2026 | 1 min read

We're setting a timeline for post-quantum cryptography migration to 2029.

Heather Adkins
VP, Security Engineering

Sophie Schmieg
Senior Staff Cryptography Engineer

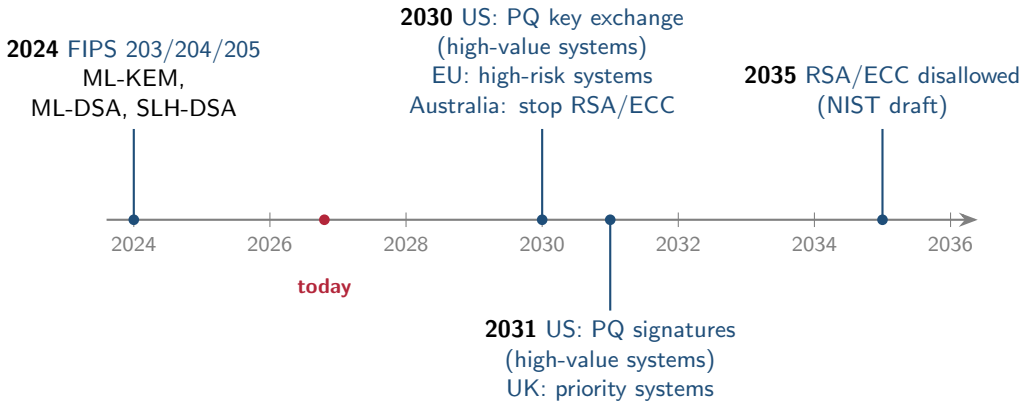


PRESIDENTIAL ACTIONS

SECURING THE NATION AGAINST ADVANCED CRYPTOGRAPHIC ATTACKS

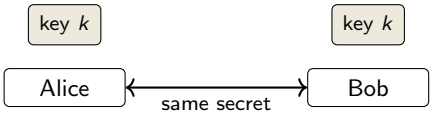
Executive Orders | June 22, 2026

Governments are deprecating pre-quantum public-key cryptography



Symmetric and public-key cryptography

Symmetric

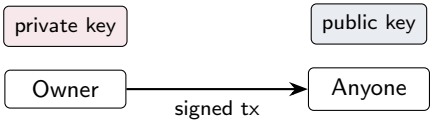


AES, SHA-256, HMAC

Fast and small

Quantum: survives

Asymmetric (public-key)



ECDSA, Schnorr, RSA

Lets strangers verify each other

Quantum: broken

Quantum Attacks

Grover

symmetric crypto

Halves the security of hashes:

SHA-256
Keccak
BLAKE
etc.

Grover, STOC 1996

Shor

asymmetric (public-key) crypto

Breaks signatures:

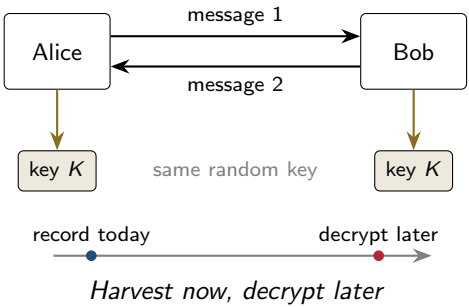
ECDSA
EdDSA
Schnorr
BLS
etc.

Shor, FOCS 1994

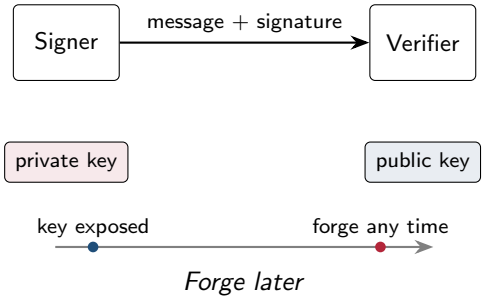
Proof-of-work holds up, but quantum attacks break signatures.

Key exchange and signatures face different deadlines

Key exchange (KEM): confidentiality



Signatures: authenticity



Web2 solved the easy half

Key exchange

- Hybrid X25519+ML-KEM-768 is the default in
  - TLS 1.3: Chrome, Firefox, iOS/macOS 26, OpenSSL 3.5
  - SSH: OpenSSH 10
- >65% of human traffic to Cloudflare is PQ-encrypted

Signatures

- ML-DSA signatures don't fit in TCP's first send, so they add a round trip to the TLS handshake
- Chrome: no PQ X.509 in its root store. *Merkle Tree Certificates* instead.

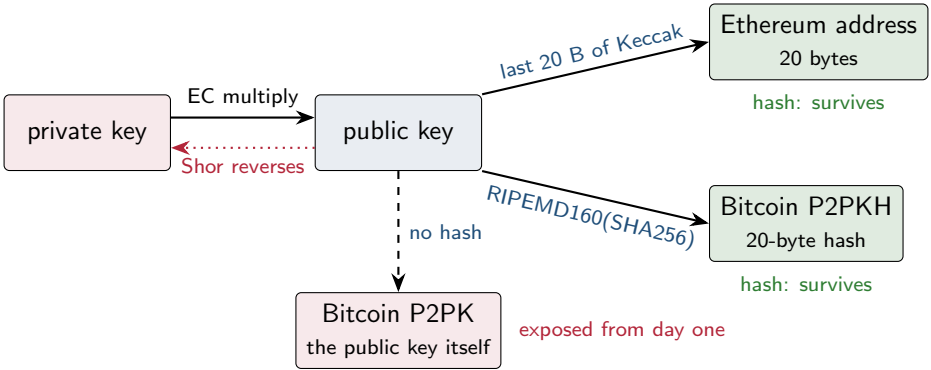
Blockchains run on signatures

What Shor breaks in today's blockchains

Primitive	**Where it is used**	**vs. quantum**
ECDSA	Bitcoin legacy, Ethereum accounts	×
Schnorr	Bitcoin Taproot	×
EdDSA	Solana, Sui, NEAR consensus	×
BLS (pairings)	Ethereum validators	×
KZG commitments	Ethereum blobs (data availability)	×
Pairing-based SNARKs	many ZK rollups	×
SHA-256, Keccak, BLAKE	mining, addresses, Merkle trees	✓
Hash-based SNARKs (e.g. STARKs)	Starknet, other rollups	✓

Shor: *infeasible* → *feasible*. Grover (vs. hashes): only a square-root speedup.

Addresses hide the public key behind a hash



A hash protects the key only until the first *spend*, which reveals the public key.

When is a key exposed?

Coin is . . .	**Public key**	**Quantum attack**
unspent at a hashed address (P2PKH; Ethereum account that never sent)	hidden	**none**: hash is quantum resistant
being spent (tx in the mempool)	visible for minutes	**race**: ∼9 min per key, ∼41% success on Bitcoin (idealized)
at a key already on-chain (P2PK, Taproot, reused address, Ethereum account that has sent)	public forever	**broken**: quantum attacker can extract the private key at their leisure

Exposed today: ∼7.1M BTC, including ∼1.7M BTC in P2PK (mostly 2009–10 coins)

Bunker mode



Justin Drake ✓
@drakefjustin

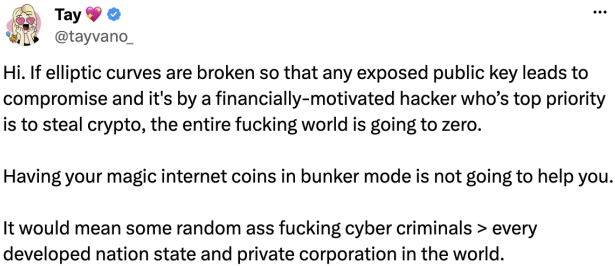
Today I call upon the blockchain industry to calmly begin planning for "bunker mode". My personal recommendation is to set in motion a controlled mass migration of assets to fresh addresses, i.e. addresses whose pubkeys remain hidden behind a hash.

Drake's worry is AI finding a *classical* attack on ECDSA, not only quantum.

Why it isn't enough

- A shared xpub reveals the "fresh" keys too
- Every spend exposes the key in the mempool
- Taproot and Solana keys are exposed by default

Who could attack after an AI break or a quantum break



Tay 💖
@tayvano_

Hi. If elliptic curves are broken so that any exposed public key leads to compromise and it's by a financially-motivated hacker who's top priority is to steal crypto, the entire fucking world is going to zero.

Having your magic internet coins in bunker mode is not going to help you.

It would mean some random ass fucking cyber criminals > every developed nation state and private corporation in the world.

AI-found classical break

anyone can attack

with "a large GPU cluster"

Early quantum computer

only **the few who own one**

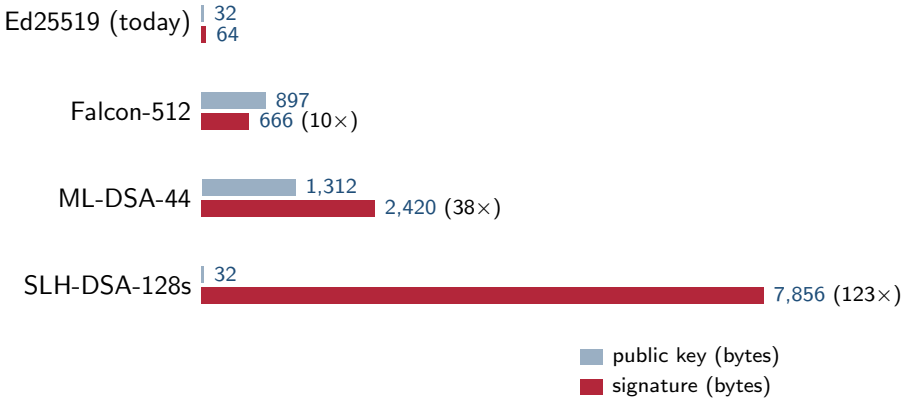
minutes per key

NIST has standardized replacements

Standard	Scheme	Job	Math	Status
FIPS 203	ML-KEM (Kyber)	key exchange	lattice	final, 2024
FIPS 204	ML-DSA (Dilithium)	signature	lattice	final, 2024
FIPS 205	SLH-DSA (SPHINCS+)	signature	hash	final, 2024
SP 800-208	XMSS, LMS (stateful)	signature	hash	final, 2020
FIPS 206	FN-DSA (Falcon)	signature	lattice	no public draft yet

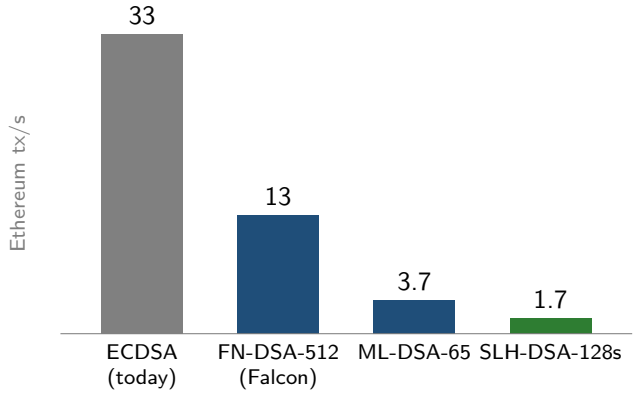
Smaller signatures may come: 9 candidates in Round 3 of NIST's additional-signature process (e.g. SQIsign $\sim$200 B, MAYO $\geq$239 B)

Standardized PQ signatures are 10–120× larger



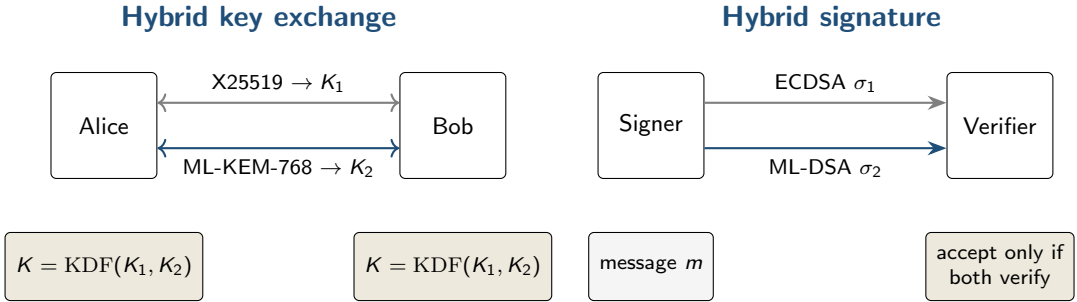
Falcon is smallest, but its floating-point signing is hard to implement safely.

Bigger signatures mean less throughput



128 recent Ethereum blocks, every signature replaced, same block size. Lattice ■, hash ■.

Hybrid schemes run classical and PQ together



Secure as long as *either* scheme holds

Blockchains are skipping hybrid

	Web key exchange	**Blockchain signatures**
Scheme	X25519 *and* ML-KEM-768	PQ scheme alone
Adding the classical half costs	+32 B each way (an X25519 key share)	+64 B signature, +32 B key per tx (Schnorr)
Relative to PQ alone	∼3% (ML-KEM-768: 1,184 B key)	∼3% (ML-DSA-44: 2,420 B signature + 1,312 B key)
Stored	seconds, then discarded	on-chain, forever

Exception: Hedera's node signing goes hybrid first, then pure FN-DSA.

Choosing between lattice and hash signatures

	Lattice (ML-DSA, Falcon)	**Hash** (SLH-DSA, XMSS)
New assumption?	yes: lattice problems	no: chains already trust hashes
If the math breaks	"whole mechanism collapses"	swap the hash function
Size	0.7–2.4 kB	3–8 kB
Algebraic structure	some (threshold, derivation)	essentially none
State to manage	none	XMSS is stateful

Hash-only designs rely on SNARK recursion, which "we don't understand . . . very well" (Thaler)

PQ signatures lose the algebra wallets rely on

Feature	Used for	EC	Lattice	Hash
Unhardened HD derivation	xpubs, watch-only wallets	✓	∼	×
MuSig / aggregation	multisig, BLS consensus	✓	∼	×
Threshold signing	MPC custody	✓	✓†	×
Silent payments	private receiving addresses	✓	∼	×

✓ works today ∼ possible, research-stage × no known construction

†threshold ML-DSA produces standard FIPS 204 signatures, but is not yet standardized or deployed

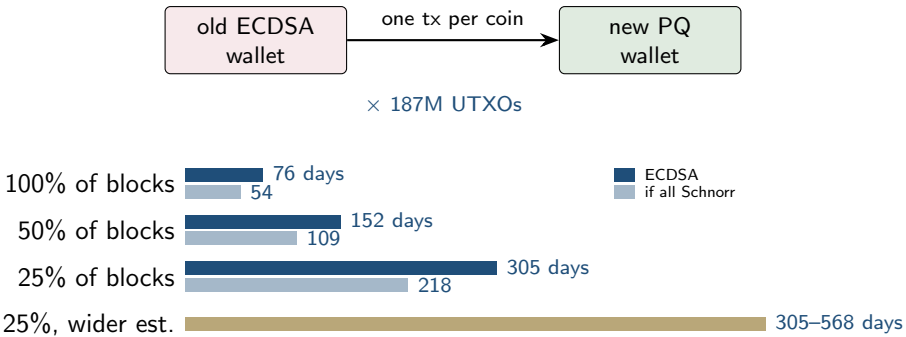


"Many of us cryptographers are actually a little sad that some of the blockchain community will move to hash-based signatures that really have no algebraic properties for us to leverage . . .

But perhaps that constraint is also where this extra sense of security comes from."

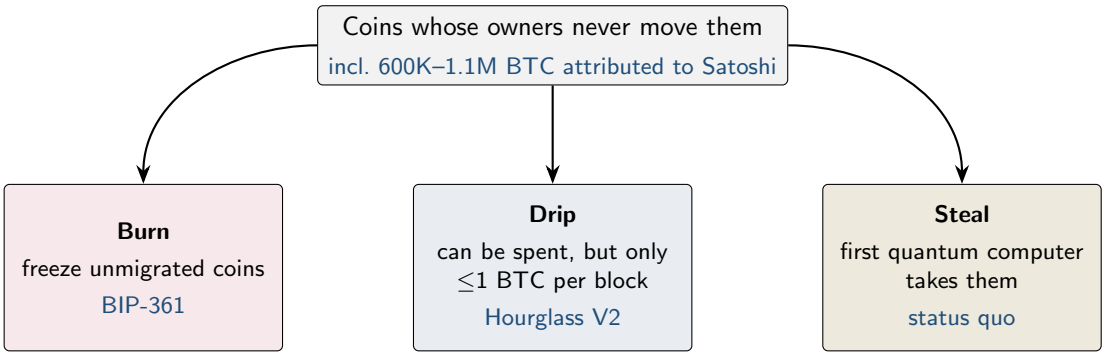
— Dan Boneh, a16z crypto, Oct 2026

Everyone has to move every coin to a new wallet



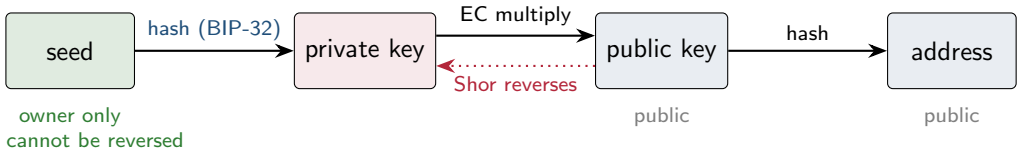
Bitcoin, idealized migration times

What happens to coins nobody moves?



A 2025 summit poll was split roughly evenly.

Proving ownership with the seed



STARK proof (hash-based, so post-quantum):
"I know a seed that hashes to the key behind this address"

The attacker has the key but not the seed. Lost seeds still cannot be rescued.
Planned by Solana (Anza), Sui, Ethereum as an emergency path.

Different challenges for consensus and execution

	Consensus (validators)	**Execution (users)**
Who signs?	hundreds of thousands of validators, every slot	millions of users, one signature per tx
Hard part	**performance** BLS aggregation has no PQ equivalent	**coordination** everyone must act; some never will
Easy part	**coordination** a known set, upgraded by a fork	**performance** a few kB per tx is survivable

Fix: **hash-based SNARK aggregation**

Fix: **account abstraction**

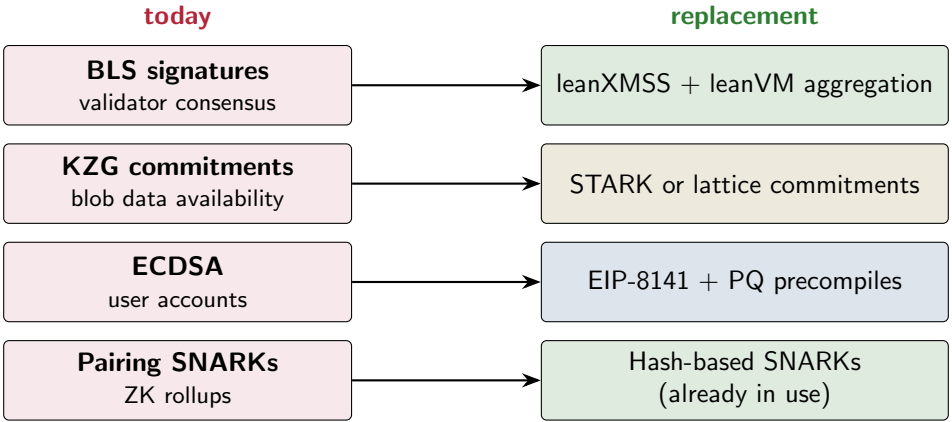
Where each chain stands

Chain	Users (EL)	Validators (CL)
Algorand	Falcon-1024 **mainnet**	state proofs only
NEAR	ML-DSA-65 **mainnet**	— (2027)
Cosmos SDK	ML-DSA-65 **shipped**	ML-DSA-65 opt-in
Starknet	Falcon-512 demo	—
Ethereum	any, EIP-8141 2027	leanXMSS devnets
Sui	ML-DSA-65 / SLH-DSA plan	—
Aptos	SLH-DSA-128s devnet	—
Tron	Falcon + ML-DSA testnet	Falcon + ML-DSA testnet
Tezos	ML-DSA-44 testnet	XMSS Quantumnet
Solana	Falcon prototype	Falcon research
Stellar	ML-DSA plan	— (TBD)
Hedera	FN-DSA 2027	FN-DSA first
XRP Ledger	ML-DSA? candidate	hybrid devnet
Bitcoin	SHRINCS or ML-DSA drafts	n/a (PoW)
Cardano	undecided	—

No chain signs consensus with a PQ scheme on mainnet yet.

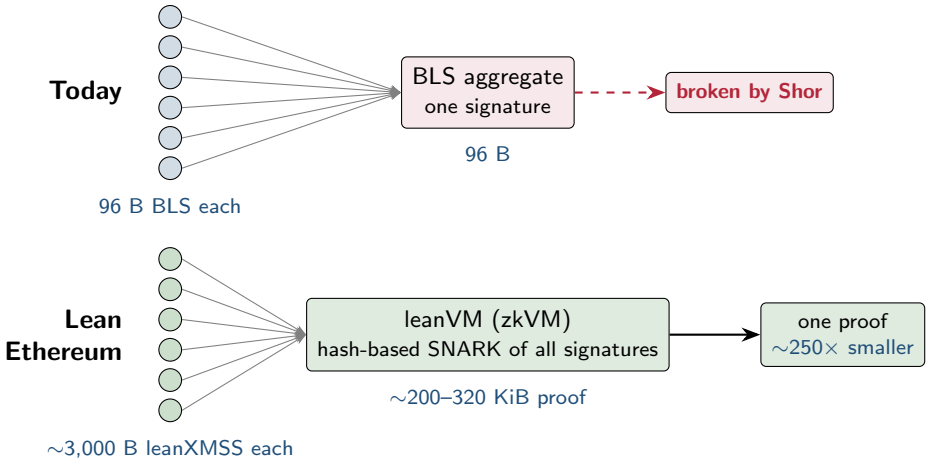
lattice: ML-DSA | lattice: Falcon | hash-based | open

Four parts of Ethereum that Shor breaks



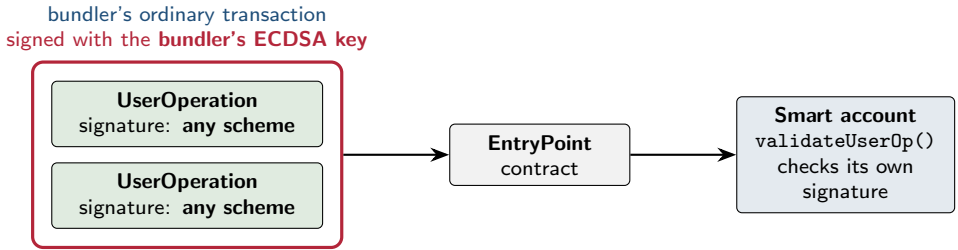
Buterin's roadmap, Feb 2026; ethereum.org

Replacing BLS in Ethereum consensus



EF Post-Quantum team (since Jan 2026), 10+ client teams on weekly PQ devnets.
leanVM is "the cryptographic cornerstone of our entire post-quantum strategy" (Drake)

ERC-4337 still wraps everything in ECDSA



Works today

- The account's code decides which signature scheme to accept, so it can be PQ

Only a partial fix

- The outer transaction is still ECDSA
- Existing accounts must move funds, or delegate via EIP-7702, which "relies on ECDSA"
- Runs outside the protocol: bundlers, a separate mempool

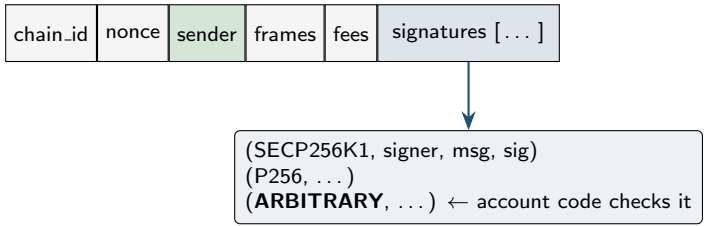
EIP-8141 drops the outer signature

Legacy (EIP-1559)



secp256k1 only
sender = ecrecover(r, s)

EIP-8141



Like ERC-4337, the account's code validates signatures. Unlike 4337 "a frame transaction carries no outer signature".

Open problems

Smaller signatures	SQIsign ($\sim$150–200 B), MAYO ($\geq$239 B), UOV (96 B, but $>$200 kB keys) in NIST Round 3
Aggregation	many hash signatures $\rightarrow$ one hash-based SNARK (Ethereum's plan)
Lost algebra	PQ HD derivation, MuSig, silent payments
Rescue proofs	ZK proof of the seed behind an exposed key
Implementable standards	fixed-point-only FN-DSA (FIPS 206), still no public draft
Dormant coins	burn, steal or drip: a governance decision

Takeaways

1. Signatures, not encryption, are the blockchain problem.

   An exposed key stays exposed forever.

2. The cryptography exists.

   The bottlenecks are block space, coordination and governance.

3. Start now.

   Migration takes years, and the deadlines are 2029–2035.